1 An electric pump is placed in a container of liquid. A model wind turbine is connected to the pump by a cable, as shown in Fig. 1.1.

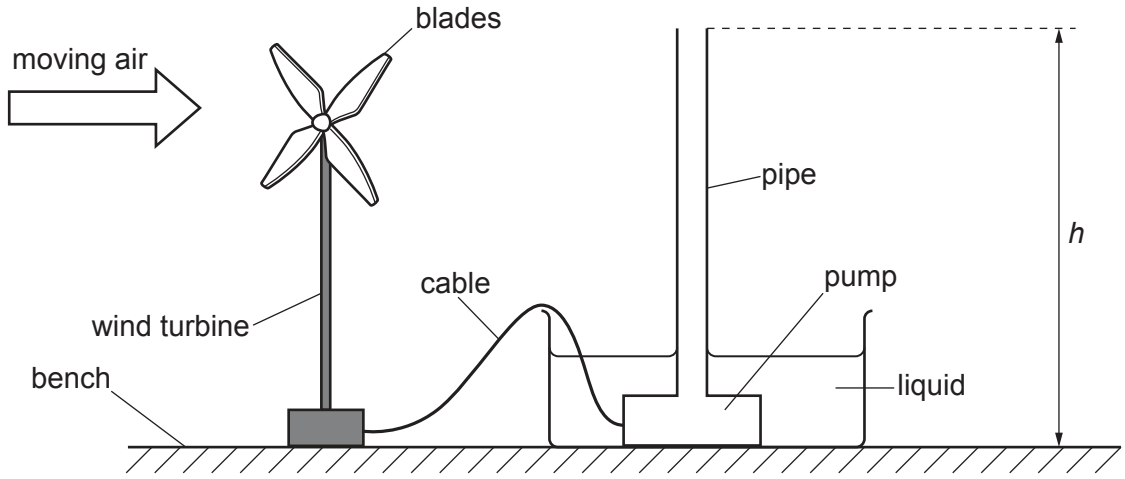

Fig. 1.1 (not to scale)

The turbine is placed in moving air. As the turbine blades turn, electricity is generated and the pump pushes liquid through a vertical pipe.

The frequency of rotation of the turbine blades is f. The height the liquid moves is h. The mass per unit time of the liquid leaving the top of the pipe is Q.

It is suggested that Q is related to f by the relationship

$$Qgh = C + Df^3$$

where g is the acceleration of free fall, and C and D are constants.

Plan a laboratory experiment to test the relationship between Q and f.

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for C and D.

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

© UCLES 2023

https://xtremepape.rs/